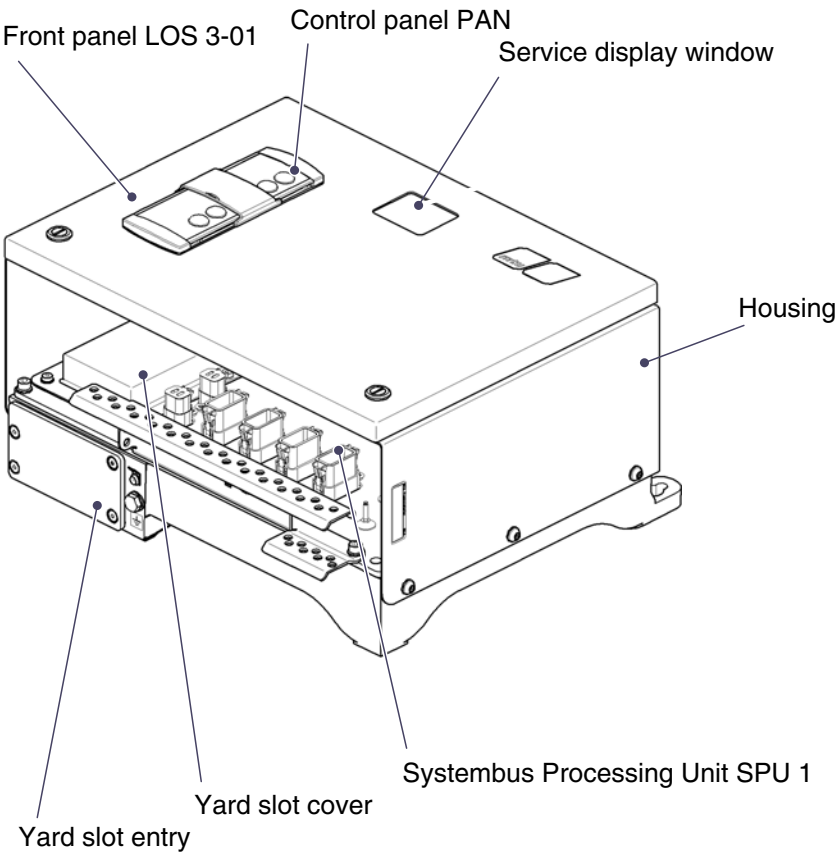
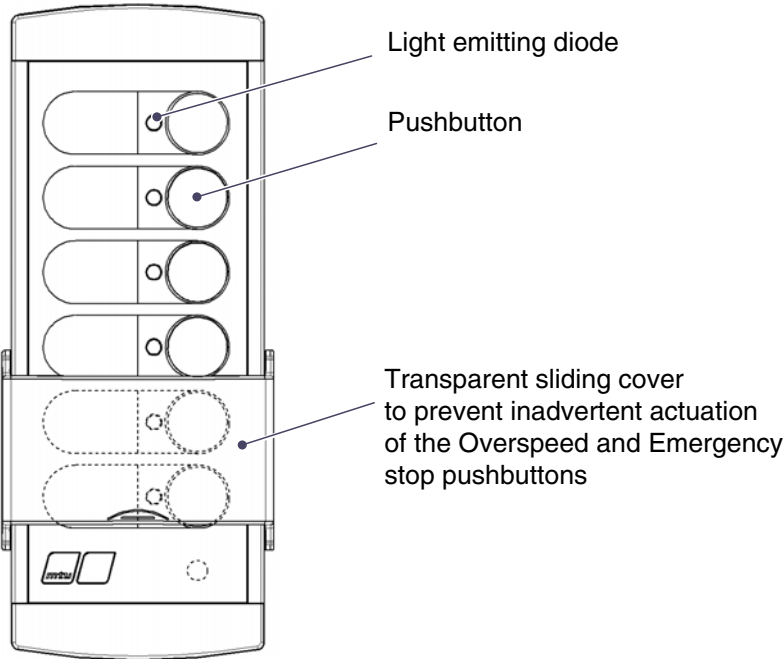


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Local Operating Panel LOP 13



Control panel LOP-PAN on Local Operating Panel LOP 13



LOP-PAN – Pushbutton functions

READY FOR OPERATION		Pressing the pushbutton switches the "Ready for operation" status on and off (manual start interlock by the operator). An active signal is a requirement to enable starting.
START		Manual start interlock by the operator. Pressing the pushbutton initiates the automatic starting sequence; only possible when READY FOR OPERATION LED is lit green.
STOP		Pressing the pushbutton initiates the stopping sequence.
COMBINED ALARM HORN OFF		Pressing the pushbutton switches the horn off.
TEST OVERSPEED		Pressing the pushbutton initiates the overspeed test.
EMERGENCY STOP		Pressing the pushbutton initiates an emergency stop.
START	STOP	Holding down the STOP pushbutton and then additionally holding down the START pushbutton cranks the engine (turning).

LOP-PAN – LED status indicators

READY FOR OPERATION		Lit	Green	"Ready for operation" status active, start possible.
		Flashing	Yellow	"Ready for operation" status active, starter not ready (cooling phase).
START		Flashing		Starting sequence started, starter not activated.
		Lit		Starter on.
STOP		Flashing		Engine running. Stopping sequence running.
		Lit		Stop command applied with engine at standstill.
COMBINED ALARM HORN OFF		Flashing	Yellow	Unacknowledged yellow alarm applied.
		Lit		Acknowledged yellow alarm applied.
		Flashing	Red	Unacknowledged red alarm applied.
		Lit		Acknowledged red alarm applied.
TEST OVERSPEED		Lit	Red	Overspeed test is active.
		Dark		Overspeed test is not active.
EMERGENCY STOP		Lit	Red	Emergency stop is active.



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Switching engine ready for operation at Local Operating Panel LOP 13



Preconditions:
Overall system is switched on.

Press READY FOR OPERATION pushbutton.

- READY FOR OPERATION LED lit.
- Engine is ready for operation.

Note: The engine plant system switched ready for operation when switched back on.

Engine start at Local Operating Panel LOP 13

Note: The automatic starting sequence can be canceled at any time by pressing the STOP pushbutton.



Preconditions:
READY FOR OPERATION pushbutton lit green.

- No start interlocks applied.



Press START pushbutton.

- START LED flashing.
- Automatic starting sequence in progress.
- START LED lit when the starter engages.
- START LED goes out when the starting sequence has been completed.
- Engine runs at idle speed

With optional Priming Pump Control (PPC):



Preconditions:
READY FOR OPERATION pushbutton lit green.
PPC selector switch set to AUTO.

- No start interlocks applied.



Press START pushbutton.

- Starting sequence as described above.
- The PPC is switched on as required
- Oil priming is canceled on pressing the START pushbutton once more (depending on EIM setting).

Stopping engine

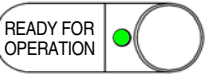


Preconditions:
Engine is running.

Press STOP pushbutton.

- STOP LED flashing.
- Automatic stopping sequence in progress.
- STOP LED lit when the engine speed is below 300 rpm.
- STOP LED goes out when the stopping sequence has been completed.
- Engine at standstill.
- Gearbox disengaged.

Engine start interlock



Preconditions:
READY FOR OPERATION LED lit green on Local Operating Panel LOP 13.

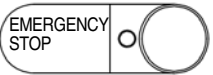
Note: The state of this function is stored when the overall system is switched off.
Engine start remains interlocked after switching back on.



Press READY FOR OPERATION pushbutton.

- READY FOR OPERATION LED goes out.
- Engine is not ready for operation.

Emergency engine shutdown



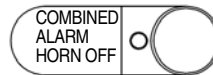
Preconditions:
Engine running.

Slide the cover into the upper locked position.

Press EMERGENCY STOP pushbutton.

- EMERGENCY STOP LED lit.
- Alarm horn sounds.
- COMBINED ALARM / HORN OFF LED flashes.
- On engines with emergency air-shutoff flaps: Flaps close.
- Engine is stopped by interrupting fuel injection.

Slide the cover into the lower locked position.



Press COMBINED ALARM / HORN OFF pushbutton.

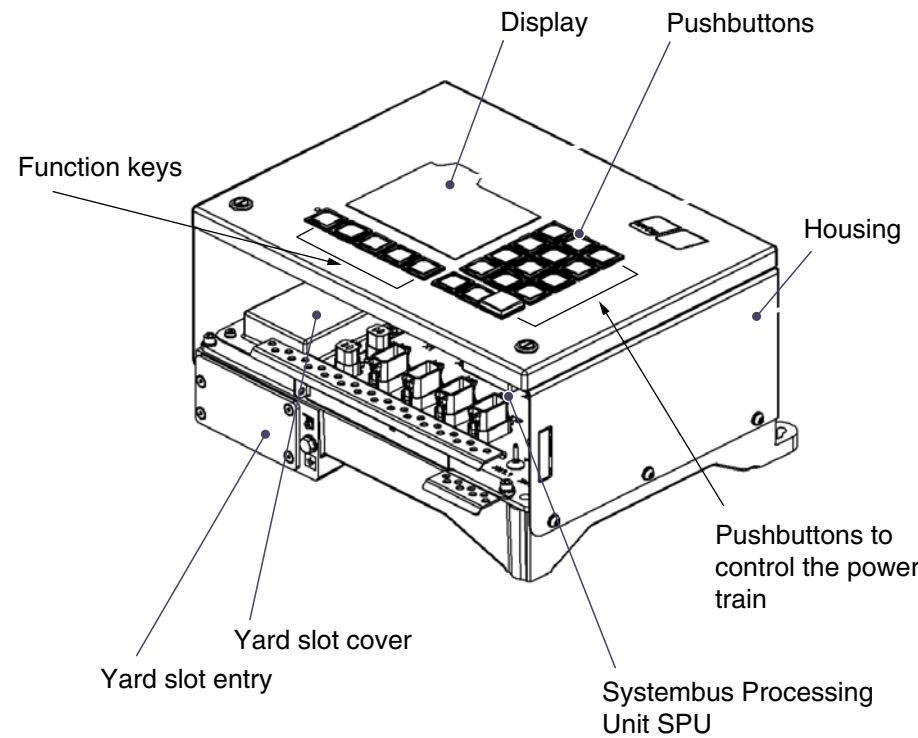
- Alarm horn is silenced.
- COMBINED ALARM / HORN OFF LED flashes.
- After acknowledging alarm at the display: COMBINED ALARM / HORN OFF LED goes out.



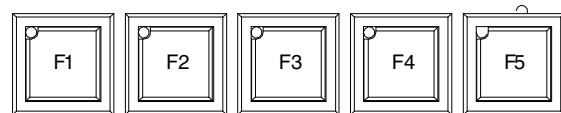
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Local Operating Panel LOP 14 (option)

The Local Operating Panel is the monitoring and control unit for the engine and gearbox*.
An alphanumeric color screen and a combined control and indicator panel are provided to facilitate operation and indication.
Measured values and alarm messages in case of limit violations are displayed in accordance with MTU specifications.

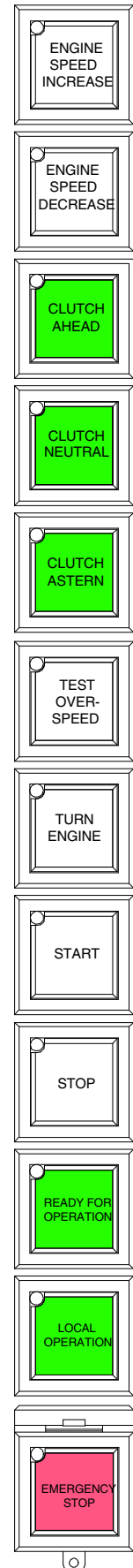


Function keys



Function keys F1 to F5 are used to control and operate the graphic screen pages on the display, to acknowledge alarms and for dimming.
The various functions of the keys are indicated by graphics and the available functions depend on which page is currently on display.
Each key is lit as long as it is held down.

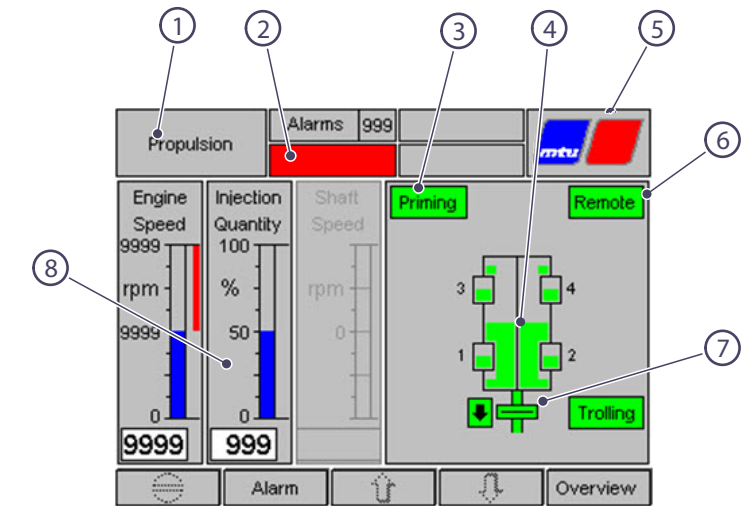
Pushbuttons to control the power train



- Holding the pushbutton down increases engine speed.
• Lit as long as the pushbutton is held down.
- Holding the pushbutton down decreases speed.
• Lit as long as the pushbutton is held down.
- Pressing the pushbutton engages the gearbox ahead.
• Flashes when AHEAD is set, but feedback for gearbox ahead is missing.
• Lit when the gearbox is engaged ahead.
- Pressing the pushbutton disengages the gearbox.
• Flashes when the NEUTRAL is set, but feedback for gearbox neutral is missing.
• Lit when the gearbox is disengaged.
- Pressing the pushbutton engages the gearbox astern.
• Flashes when the ASTERN is set, but feedback for gearbox astern is missing.
• Lit when the gearbox is engaged astern.
- Pressing the pushbutton initiates an overspeed test.
• Lit during the overspeed test.
- Pressing the pushbutton initiates engine cranking.
• Lit during the cranking sequence.
- Pressing the pushbutton initiates engine start.
• Lit during the starting sequence.
• Lit when the TURN pushbutton is actuated together with the STOP pushbutton.
- Pressing the pushbutton initiates engine shutdown.
• Lit during the stopping sequence.
• Lit when the TURN pushbutton is actuated together with the START pushbutton.
- Pressing the pushbutton switches the power train ready for operation.
The operator thus cancels the manual start interlock.
• Lit when the "Ready for operation" status is active.
- Pressing the pushbutton activates Local mode.
• Lit when Local mode is activated.
- Pressing the pushbutton initiates an emergency engine stop.
• Lit when an emergency engine stop has been initiated.
• Lit until the "Emergency engine stop" has been acknowledged.

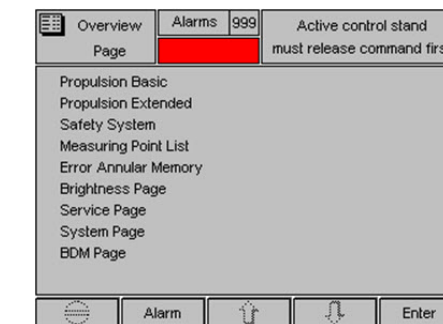
* Optional/depending on plant configuration

Screen pages on Local Operating Panel LOP 14

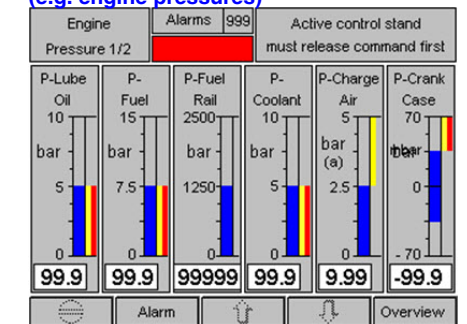


- 1 Screen page name
- 2 Active alarm indicator
- 3 Oil priming active
- 4 Engine and exhaust turbocharger indicator
- 5 Message window
- 6 Local/Remote control active
- 7 Gearbox status/Trolling active
- 8 Measuring point bargraph display

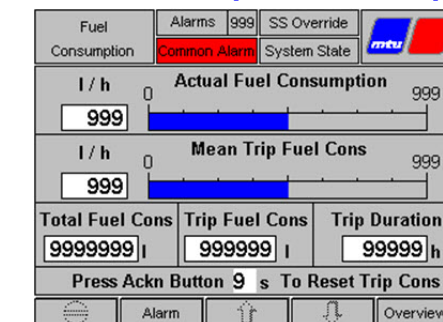
Overview page



Engine screen page (e.g. engine pressures)



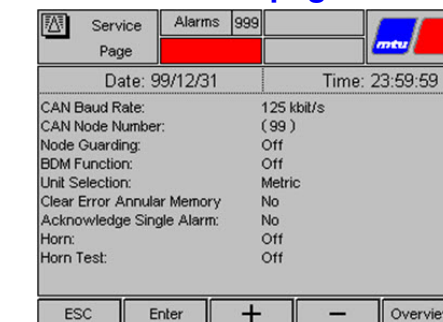
Fuel consumption screen page



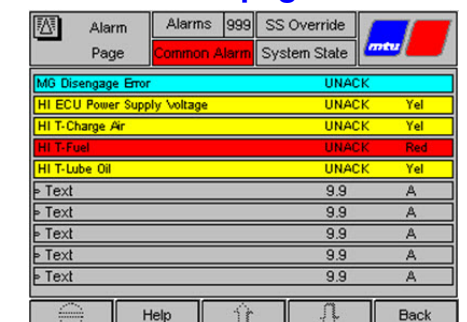
Brightness screen page



Service screen page



Alarm screen page

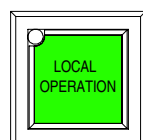




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Local mode

Activating:

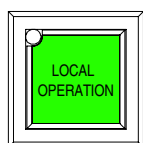


Precondition:
LOCAL
OPERATION
pushbutton is dark.

Press key.

Pushbutton lit.
Local mode activated:
Power train can be controlled by LOP.
Power train can no longer be
controlled via RCS and MCS
(Exception: Emergency stop)

Deactivating:



Press key.

Pushbutton goes out.
Local mode deactivated:
Power train can no longer be
controlled by the LOP.
Power train can be controlled via
RCS and MCS.

Switching power train ready for operation (canceling manual start interlock)



Precondition:
READY FOR
OPERATION
pushbutton dark
(i.e. manual start
interlock is activated).

When the power
train is ready for
operation:
Press pushbutton.

Pushbutton lit.
Start interlock is canceled.

Engine start interlock

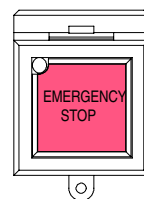


Precondition:
READY FOR
OPERATION
pushbutton lit
(i.e. manual start inter-
lock is not activated).

To prevent engine
start for the power
train:
Press pushbutton.

Pushbutton goes out.
Start interlock is activated.

Emergency engine stop



Precondition:
EMERGENCY STOP
pushbutton is dark.

Lift the protective flap
and press pushbutton
EMERGENCY STOP.

Pushbutton lit.
Alarm signaling at the control
stands (e.g. horn).
Engine is emergency stopped
(by closing emergency air-shutoff
flaps or interrupting fuel supply).

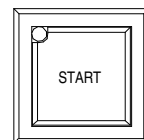
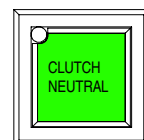
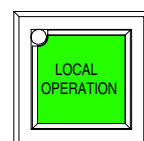
Silence horns at control
stands.
(MCS-6).

Alarm signaling at the control
stands stops.

Alarm acknowledgment
at main control stand
(MCS-6).

“Emergency stop” alarm is
acknowledged.

Engine start



Preconditions:
LOCAL OPERATION
and READY FOR
OPERATION
pushbuttons lit.

With optional PPC:
PPC selector switch set to AUTO.

CLUTCH/NEUTRAL
pushbutton lit.

START pushbutton
dark.

Press pushbutton.

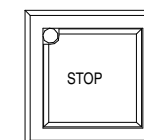
START pushbutton lit.
Automatic starting sequence.
The PPC is switched on as
required (oil priming does not take
place when starting is repeated).
On completion of the starting
sequence:
Engine runs at idle speed.
Pushbutton goes out.

Oil priming is canceled on pressing
the START pushbutton once more
(depending on EIM setting)

Engine shutdown



Precondition:
Engine running.
LOCAL OPERATION
pushbutton lit.

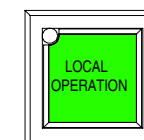


STOP pushbutton
dark.

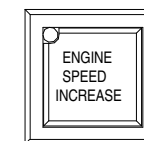
Press pushbutton.

Pushbutton lit. Automatic stopping
sequence. On completion of the
stopping sequence: Engine is
shut down; pushbutton goes out.

Changing engine speed



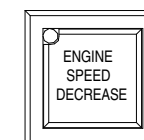
Precondition:
Engine running.
LOCAL OPERATION
pushbutton lit.



ENGINE SPEED
pushbutton dark.

Press and hold down
the pushbutton.

Pushbutton is lit as long as it is
held down.
Engine speed increases.



ENGINE SPEED
pushbutton dark.

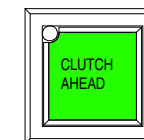
Press and hold down
the pushbutton.

Pushbutton lit as long as it is
held down.
Engine speed decreases.

Changing gearbox state (reversing gear)



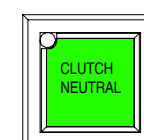
Precondition:
Engine running.
LOCAL OPERATION
pushbutton lit.



CLUTCH AHEAD
pushbutton dark.

Press pushbutton.

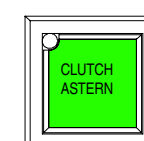
Engine speed in engagement
window. Pushbutton lit.
Gearbox is engaged ahead.



CLUTCH NEUTRAL
pushbutton dark.

Press pushbutton.

Engine speed in engagement
window. Pushbutton lit.
Gearbox is disengaged.



CLUTCH NEUTRAL
pushbutton dark.

Press pushbutton.

Engine speed in engagement
window. Pushbutton lit.
Gearbox is engaged astern.



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Control panel Start/Stop PAN



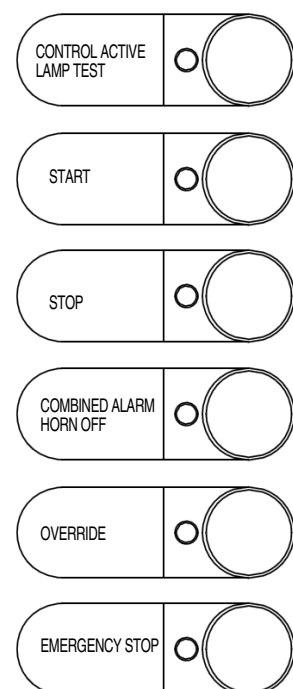
Light emitting diode

Pushbutton

Transparent sliding cover to prevent inadvertent actuation of the Safety system override and Emergency stop pushbuttons.

Controls on control panel Start/Stop PAN

Pushbuttons:



Pressing the pushbutton initiates a lamp test.

Pressing the pushbutton initiates the automatic starting sequence; only possible when CONTROL ACTIVE/LAMP TEST LED is lit green.

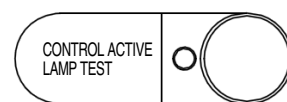
Pressing the pushbutton initiates the stopping sequence.

Pressing the pushbutton switches the horn off.

Pressing the pushbutton overrides the engine safety system.

Pressing the pushbutton initiates an emergency stop.

Indicators on control panel Start/Stop PAN

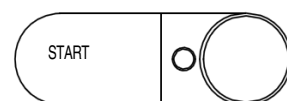


Lit

Lit

Control stand active. Start possible.

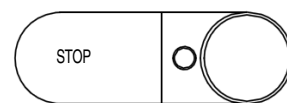
Control stand active. Start not possible, start interlock applied.



Flashing

Lit

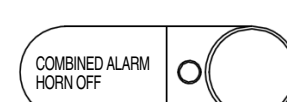
Engine starting sequence running.
Engine running.



Flashing

Lit

Stopping sequence running.
Stop command applied with engine at standstill.



Flashing

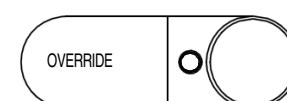
Lit

Unacknowledged yellow alarm applied.
Acknowledged yellow alarm applied.

Flashing

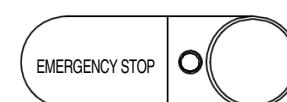
Lit

Unacknowledged red alarm applied
Acknowledged red alarm applied.



Lit

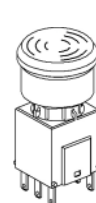
Safety system override is active.



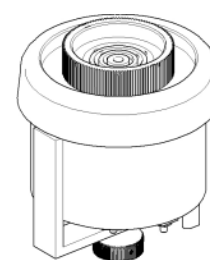
Lit

Emergency stop is active.

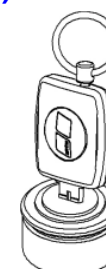
Key switch (Option) and alarm horn (Option)



Alarm horn for enclosed control stands
Ø 28mm/87dB/IP66



Alarm horn for exposed control stands
Ø 52mm/90dB/IP55



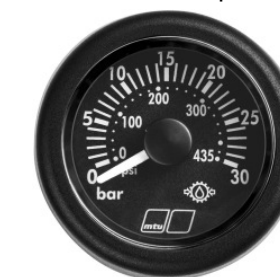
- Main key-operated switch (Master-Key-Switch)
- RCS key-operated switch (RCS-Key-Switch)

Analog display instruments

Gear oil temperature



Gear control oil pressure



Engine speed
(Speed range depends on engine)



Engine lube oil pressure



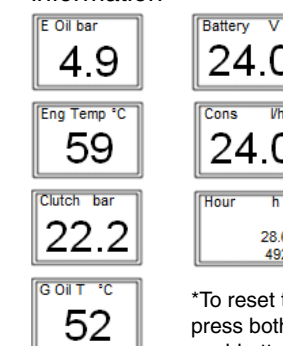
Engine coolant temperature



Pushbuttons Display

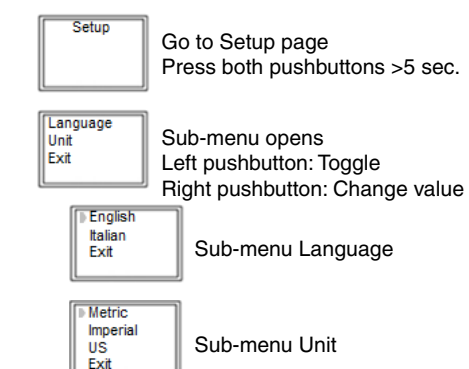
Speed instrument display

Display shows additional information

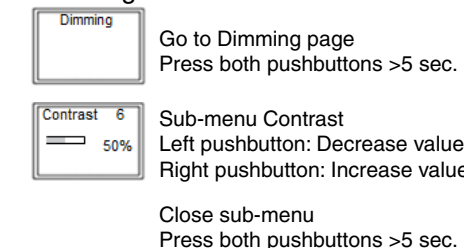


*To reset trip time press both pushbuttons >5 sec.

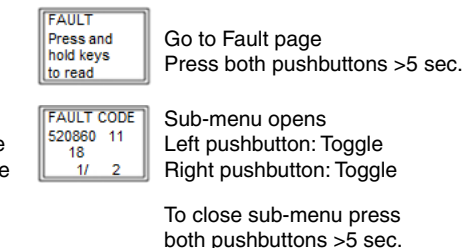
Setup Display



Dimming



FAULT



Contents: SPN
OC occurrence counter
Alarm no.
FMI
Number of alarms



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Switching on the overall system, without key switch

Preconditions:
Overall system is switched on with master switch (circuit breaker).

Switch on master switch (circuit breaker) for the power train.

Note:
Remote/Local operating status is stored.
When started, the plant changes automatically to the stored operating status.

- Local Operating Panel LOP of the power train is booted.
- Background illumination of MCS and RCS of the power train lights up.
- Utility lamp of the Start/Stop PAN lights up for the power train concerned.
- Command (CMD) LED on the command units on the control stands of the power train lit (Remote mode). The LED is lit if the plant was in Local mode when last switched off.
- All LEDs on the Start/Stop PAN and the command units lit: Automatic lamp test in progress.

Repeat procedure for other power trains (if applicable). Procedures are repeated in the same way as for the first power train.

Plant with LOP 14:
If the plant was previously not in Local mode when switched off:
• COMMAND LED flashes on those control stands capable of assuming initial command.
• Command unit buzzers sound (providing that RCS Quiet mode is deactivated)

Plant with LOP 13:
• COMMAND LED flashes on those control stands capable of assuming initial command.
• Command unit buzzers sound (providing that RCS Quiet mode is deactivated).

Switching on the overall system, with master/RCS key-operated switch (optional)

Preconditions:
Overall system is switched on with master switch (circuit breaker).



Insert the master/RCS key in the key switch on the main control stand and turn clockwise.

Note:
Remote/Local operating status is stored. When started, the plant changes automatically to the stored operating status.

- Local Operating Panel LOP of the power train is booted
- All Local Operating Panels are automatically booted in Standby mode on multiple-engine system (if applicable).
- Background illumination of MCS and RCS of the power train lights up.
- Utility lamp of the Start/Stop PAN lights up for the power train concerned.
- CMD LED on the command units on the control stands of the power train flashes (Remote mode). The LED is lit if the plant was in Local mode when last switched off.
- The buzzers of the RCS sound.

Repeat procedure for other power trains (if applicable). Procedures are repeated in the same way as for the first power train.

Switching off the overall system, without key switch

Preconditions:
Overall system of the power train is switched on. Engine at standstill.

Switch off master switch (circuit breaker) for the power train.

- Overall system of the power train is out of operation.

Repeat procedure for other power trains (if applicable). Procedures are repeated in the same way as for the first power train.

Switching off the overall system, with master/RCS key-operated switch (optional)

Preconditions:
On-board power supply is switched on. Engine at standstill.



Insert the master/RCS key in the key switch on the main control stand and turn counterclockwise.

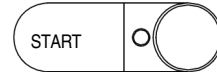
- Overall power train system is out of operation.
On multi-engine plants, all Local Operating Panels remain in Network-ON (Standby mode), until the last power train has been switched off.

Repeat procedure for other power trains (if applicable). Procedures are repeated in the same way as for the first power train.

Engine startup at the Start/Stop PAN

Preconditions:
• Overall system of the power train is switched on.
• Propulsion control lever on the command unit of the commanding control stand is set to NEUTRAL.
• No engagement request and no start interlocks applied.
• LED CONTROL ACTIVE/LAMP TEST on control panel Start/Stop PAN on the control stand lights up green.

Note: The automatic starting sequence can be canceled at any time by pressing pushbutton STOP.



Press START button.

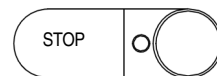
Note:
Starting is not executed if CONTROL ACTIVE/LAMP TEST is yellow when the Start pushbutton is actuated. An appropriate message indicating why starting cannot be executed appears on the display of the control stand.

- LED START flashes while the automatic starting sequence is executed.
- LED START lights up.
- For optional oil priming pump: Engine is primed.
- Engine speed increases.
- Engine runs at idle speed on completion of the starting sequence.
- Power train is ready for engagement.

Repeat procedure for each additional engine (if applicable). Starting sequence is repeated in the same way as for the first engine.

Engine shutdown at the Start/Stop PAN

Preconditions:
• Propulsion control lever of commanding control stand should be set to NEUTRAL.
• LED CONTROL ACTIVE/LAMP TEST on control panel Start/Stop PAN on the control stand lights up green.
• Engine running at idling speed.



Press STOP pushbutton.

- LED STOP flashes while the automatic stopping sequence is carried out.
- Engine speed decreases.
- LED STOP lights up when the engine speed has fallen below 300 rpm.
- The engine is at a standstill when the automatic stopping sequence has been completed and LED STOP goes out.

Repeat procedure for each additional engine (if applicable). The stopping sequence is repeated in the same way as for the first engine.

Emergency engine shutdown at the Start/Stop PAN

Preconditions: Engine running.

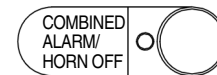


Slide the locking device up until it engages.

Press EMERGENCY STOP pushbutton.

- LED EMERGENCY STOP lights up.
- Alarm horn sounds.
- LED COMBINED ALARM / HORN OFF flashes up red.
- On engines with emergency air-shutoff flaps: Flaps close.
- Engine is shut down by stopping fuel injection.

Slide the locking device down until it engages.



Press COMBINED ALARM / HORN OFF pushbutton.

- The alarm horn is silenced.
- LED COMBINED ALARM / HORN OFF flashes up red.
- After acknowledging alarm at display: LED COMBINED ALARM / HORN OFF goes out.

Activating/deactivating System Override at Start/Stop PAN

When the Override function is active, the overall system ignores certain operating states which would usually lead to automatic power reduction or engine shutdown in normal operation:

- Note:
- MTU refuses to accept warranty claims for engine damage incurred while the Override function is active! Activation of the Override function is stored in the Engine Control Unit.
 - With the engine at a standstill: Start interlocks are ignored, e.g. the engine can be started even with the gearbox engaged.
 - With the engine running: Engine operating limit value violations no longer lead to power reduction or engine shutdown.
Exception: Violation of the overspeed limit value still leads to shutdown.

Activating override:



Slide the locking device up until it engages.

Press OVERRIDE pushbutton.

- LED OVERRIDE lights up.
- Alarm horn sounds.
- LED COMBINED ALARM / HORN OFF flashes up red.

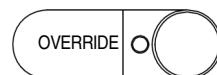
Slide the locking device down until it engages.



Press COMBINED ALARM / HORN OFF pushbutton.

- The alarm horn is silenced.
- LED COMBINED ALARM / HORN OFF flashes up red.
- Engine runs in Override mode.
- Rectify any malfunctions and alarm messages manually.
- After acknowledging alarm at display: LED COMBINED ALARM / HORN OFF lights up.

Deactivating override:



Slide the locking device up until it engages.

Press OVERRIDE pushbutton.

- LED OVERRIDE goes out.
- Operating mode Override is deactivated.
- LED COMBINED ALARM / HORN OFF goes out.

Slide the locking device down until it engages.



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Other screen pages

Displaying Help for key assignment



Press key (9) "Menu".

Note: The dialog window closes automatically after 5 sec.



Changing to Day/Night view



Press key (7).



Adjusting display/instrument brightness



Press key (7) "Day/Night" for longer than 2 sec.



Adjust the brightness with keys (2) and (4).

Adjust the instruments with keys (6) and (8).

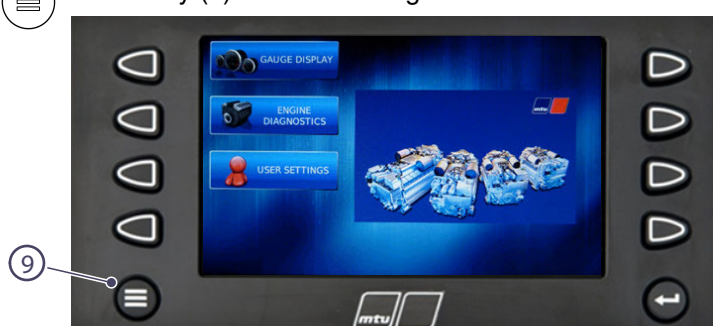
Note: The keys are also dimmed.

Press again to reset the dimming function key (7) Day/Night.

Opening the Main menu



Press key (9) "Menu" for longer than 2 sec.



Main menu User settings



Press key (5) "User settings" in the Main menu. Menu for user settings opens.



Select the settings with keys (2) Up and (4) Down.

Select the settings with keys (8) Down and (6) Up.

Save the values with key (7) and

confirm the changed values with key (10) "Enter".

Return to the Main menu with key (9) "Menu".

Close the Main menu with key (1).

Opening the Alarm page



Press key (3) "Alarm overview".

Note: The Alarm page only opens when an alarm is pending acknowledgment.



Switch off the horn by pressing key (1) "Horn Off".

Scroll through the Alarm list with key (6) "Up" and (8) "Down".

Confirm the alarm with key (10) "Enter".

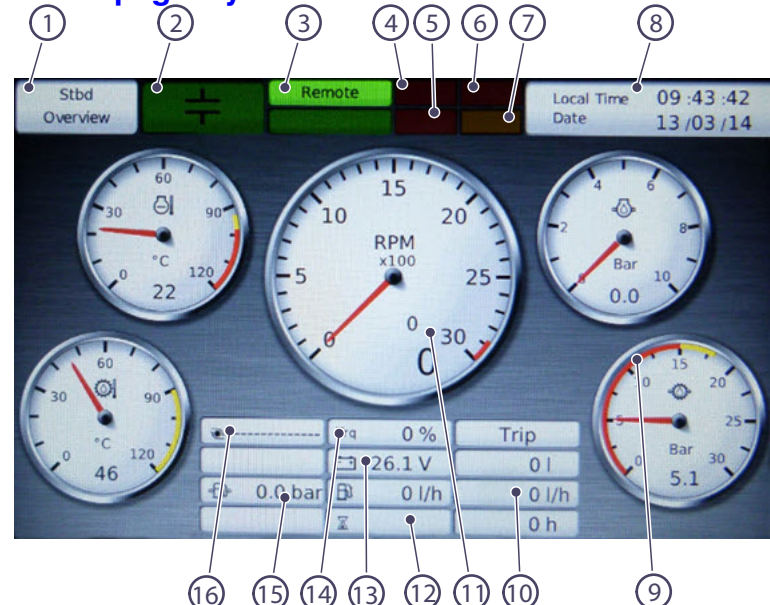
Opening the measuring-point list



Open the page with keys (2) Up and (4) Down.



Screen page layout



- | | |
|--------------------------------|--|
| 1 Screen page name | 8 Display of date/time or active alarm |
| 2 Gearbox state | 9 Display of limit values |
| 3 Display of Local/Remote mode | 10 Fuel consumption |
| 4 Override active | 11 Speed demand |
| 5 Power reduction active | 12 Operating hours |
| 6 Red alarm active | 13 Battery voltage |
| 7 Yellow alarm active | 14 Engine load |
| | 15 Fuel pressure after filter |
| | 16 Exhaust turbocharger active |



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MFD display key assignment

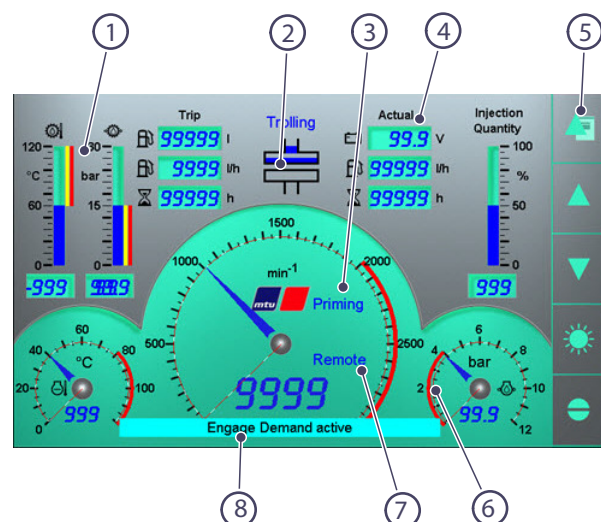
Function keys (1 to 5)

For navigation in the menu and selection of parameters.



- 1 Open/close the Alarm page
- 2 Scroll one page up
- 3 Scroll one page down
- 2/3 Hold down both keys --> Open the Overview page
- 4 Day/Night mode
- 5 Alarm acknowledgment

MFD display screen layout



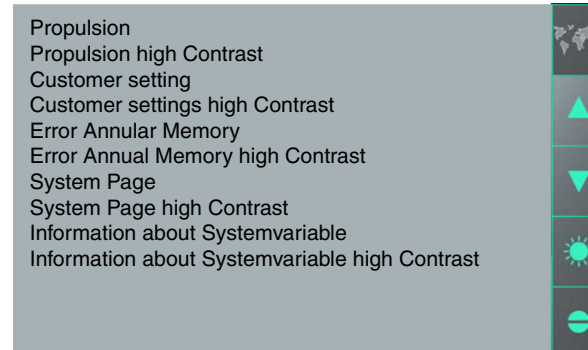
- 1 Gear oil temperature/oil pressure
- 2 Gearbox state/Trolling active
- 3 Oil priming active
- 4 Battery voltage
- 5 Alarm indication (yellow and red alarm)
- 6 Limit value indication
- 7 Remote/Local mode indication
- 8 Message window

Sub-menus

Sub-menus

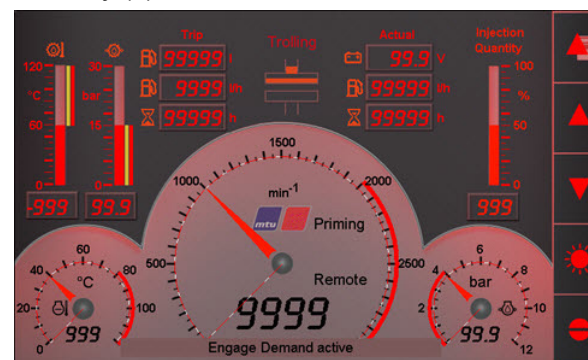
- | | | | | | |
|---|--|---------------------|---|--|-------------------|
| ① | | Change language | ④ | | Scroll up |
| ② | | Change input values | ⑤ | | and down |
| ③ | | | ⑤ | | Confirm selection |

Overview page



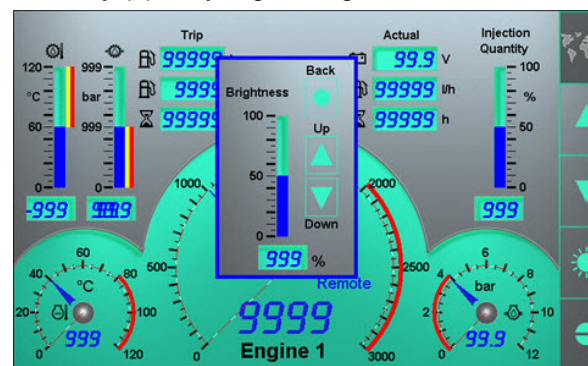
Changing to Day/Night view

Press key (4).

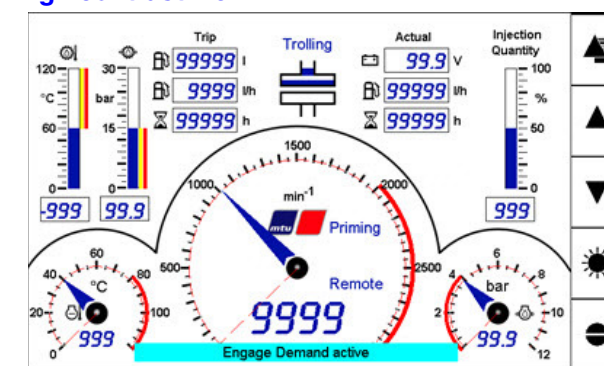


Adjusting display/instrument brightness

Press key (4) "Day/Night" longer than 2 sec.



High contrast view



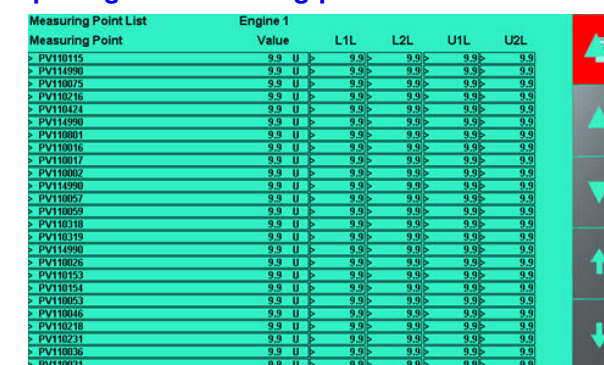
Opening the Alarm page

Press key (1).

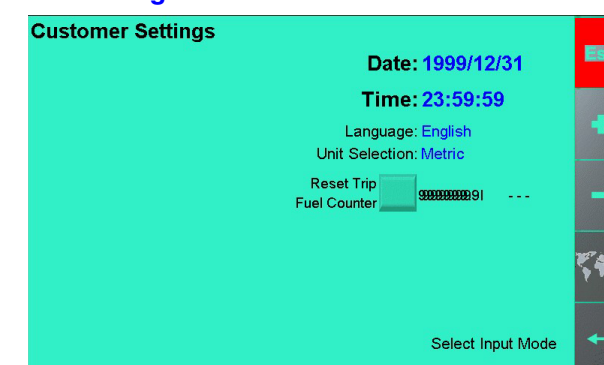


- Flashing yellow: Unacknowledged yellow alarm
- Flashing red: Unacknowledged red alarm

Opening the measuring-point list



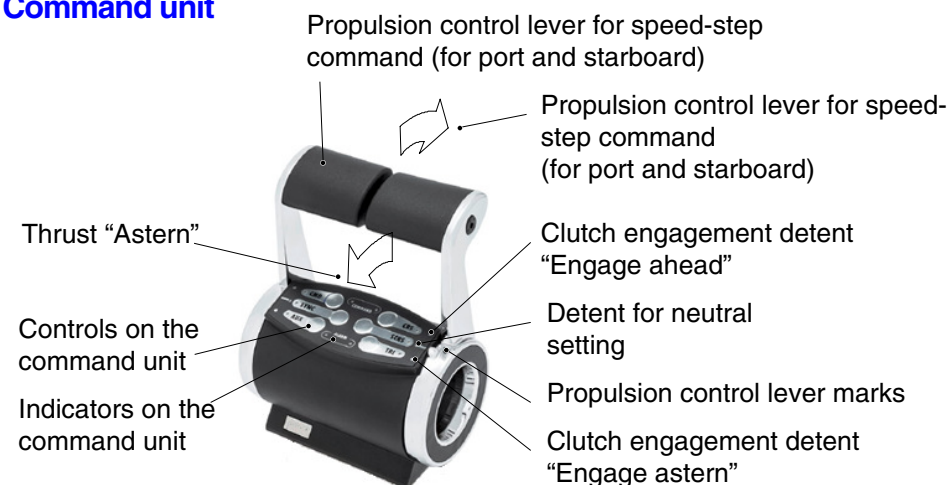
User settings





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Command unit



Controls on the command unit

CMD

- Key to
 - Select the commanding control stand
 - Acknowledge RCS alarms
 - Activate the WARM UP function

SYNC

Key to activate the SYNC or SMART SYNC. function.

CRS

Key to activate CRUISING mode.

AUX

Key to activate an auxiliary function defined in the system configuration.

SCRS

Key to activate SMART CRUISING mode.

TRL

Key to activate TROLLING mode.

Indicators on the command unit



Two LEDs to indicate which command unit is in command (separate for engine 2 - port and engine 1 - starboard).



Two LEDs to indicate a combined alarm from Remote Control System RCS-6 (separate for engine 1 and engine 2).



LED to indicate that the key function is activated.

Operating the command unit



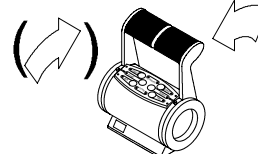
Engaging ahead (astern):
Move the propulsion control lever forwards (back) out of the NEUTRAL position to the engagement lock.



Increasing speed:
Move the propulsion control lever further forward (or further back if engaged astern).



Decreasing speed:
Move the propulsion control lever further back (or further forward if engaged astern).



Disengaging:
Lock propulsion control lever in NEUTRAL.

Assuming initial command

At the control stand which is to assume initial command:



Set both propulsion control levers to NEUTRAL.



Press CMD key.

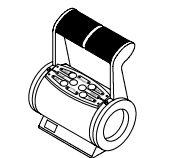


Commanding control stand:
COMMAND LEDs lit; horns silenced.

Inactive control stands:
COMMAND LEDs dark; horns silenced.

Command transfer

Assuming command at another control stand:
Commanding control stand:



Both propulsion control levers in any position.

Control stand at which command is to be assumed:



Set both propulsion control levers to NEUTRAL.

Press CMD key.

Set both propulsion control levers to any position.



Previously commanding control stand:
COMMAND LEDs go out. Horn sounds for 2 sec.

COMMAND LEDs lit. Speed demand and gearbox state are set by the input signals from the newly selected control stand after 2 sec.

Warm-up (changing speed without engagement)

Activating warm-up:
Commanding control stand:



Preconditions:
Set propulsion control lever to NEUTRAL.



Press and hold down CMD key on propulsion control lever.



Set propulsion control lever to desired engine speed.

Release CMD key.

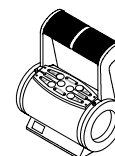


COMMAND LED flashing.

Gearbox remains in neutral.
Engine speed increases.

Engine speed changes in accordance with propulsion control lever setting.

Deactivating warm-up:
Commanding control stand:



Set propulsion control lever to NEUTRAL.



Press CMD key on propulsion control lever.



COMMAND LED lit.
Gearbox is in neutral.
Engine runs at idle speed.

Acknowledging a combined alarm horn

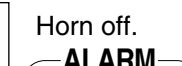
Commanding control stand:



State:
Horn sounds.
ALARM LED flashes for affected power train.



Press CMD key.



Horn off.
ALARM LED continues flashing until the alarm is acknowledged at a display.





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Changing propulsion mode

Commanding control stand:

All propulsion control levers are in NEUTRAL (also for propulsion plants with three or four shafts).
Propulsion mode CRS (Cruising Mode) is active after switching on the overall system.



Set to NEUTRAL if necessary.



TRL LED lit on all control stands.



Example, Trolling:
Press TRL key.

Single Control Lever mode

Several modes are possible depending on the basic configuration of the plant:

- Single Control Lever mode (SCL)
- Synchronization mode (Smart SYNC)

Activating Single Control Lever mode (in Single Control Lever mode)

Commanding control stand:



Preconditions:
Propulsion control lever not set to NEUTRAL.



Press SYNC key.



All except for the propulsion control lever of engine 1 (SCL Master)
Set to NEUTRAL within 4 sec.

Propulsion control lever of engine 1/shaft 1 is in Single Control Lever mode of the master control lever (SCL Master) which controls all other engines/shafts (SCL Slave).



COMMAND LED lit on Master control lever.



SYNC LED lit on all control stands.
The propulsion control lever of engine1 / shaft 1 (SCL Master) now controls all engines/shafts.

Deactivating Single Control Lever mode (in Single Control Lever mode)

Commanding control stand – to remain engaged:
Deactivation by key (“SYNC key actuation” setting)



Move the propulsion control lever of the SCL Slave shaft out of NEUTRAL and match with the Master control lever (+/- 20%).



Press SYNC key.



SYNC LED flashes on all control stands.
Horn (SCL Slave shaft) sounds.



COMMAND LEDs lit for both power trains.



SYNC LED dark on all control stands.
Horn (SCL Slave shaft) silenced.
Each propulsion control lever controls the associated power train.

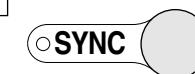
Deactivating with the slave propulsion control lever from neutral (“Panic” setting)



Move the propulsion control lever of the SCL Slave shaft out of NEUTRAL.



COMMAND LED lit for both power trains.



SYNC LED flashes blue and goes out on all control stands.
Each propulsion control lever controls the associated power train.

Commanding control stand – to remain disengaged:

Deactivating with Master control lever in neutral (“Master not ahead” setting)



Preconditions:
Propulsion control lever of the SCL Master shaft not set to NEUTRAL.



COMMAND LED lit again for both power trains.



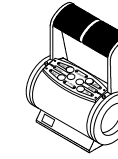
SYNC LED flashes blue and goes out on all control stands.
Horn (SCL Slave shaft) silenced.
Each propulsion control lever controls the associated power train.



Lock the propulsion control lever of the SCL Master shaft in NEUTRAL.

Activating Synchronization mode (in SMART SYNC mode)

Commanding control stand



Preconditions:
All power trains are engaged in the same direction.

If propulsion control lever settings match



Press SYNC key.



The COMMAND LED associated with the Master control lever remains activated.
The COMMAND LED associated with the Slave control lever is deactivated.
The speed of all engines is adjusted to the input signal from the Master control lever.



Control gearbox and speed of all engines with the Master control lever.



SYNC LED lit on all control stands.
All engines run at the same speed.

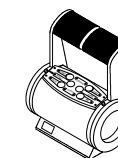
If propulsion control lever settings do not match



Press SYNC key.



The SYNC LED is activated on all command units on all control stands.



Set propulsion control lever to same position.

Engine speed is synchronized.
The COMMAND LED of the Master control lever remains activated.
The COMMAND LED of the Slave control lever is deactivated.

Deactivating Synchronization mode (in SMART SYNC mode)

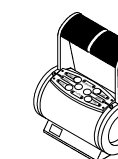
Commanding control stand



Press SYNC key.



The SYNC LED is deactivated on all command units on all control stands.



Control gearbox and engine speed separately for all engines.

Note: MTU engine numbering system
Engine 1 = Starboard (outer)
Engine 2 = Port (outer)
Engine 3 = Starboard (inner) or Center
Engine 4 = Port (inner)